

Plan Audit Certificates as the RPLAN/RCTX Fixed Point

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Abstract

This draft presents RPLAN, RCTX, audit certificates, manifests, and verifier integration as the BISECT fixed point for heterogeneous redistricting algorithms. It is an audit/certification infrastructure paper: certificates verify declared profiles, canonical ordering, hashes, package integrity, and lineage, not political fairness or legal compliance beyond the recorded profile.

1 Introduction

U.20 documents ‘rplan-core’, ‘rplan-io’, ‘rplan-audit’, ‘bisect verify’, and ‘rplan-cli’. The paper asks whether BISECT can use a neutral plan/context/certificate family as the fixed point for construction, exact, local-search, Pareto, and future algorithm families.

2 Schema Boundary

RPLAN stores plan identity and assignments. RCTX stores graph, population, canonical unit order, and context needed for verification. Certificates and manifests record declared profiles, verifier results, hashes, lineage, and algorithm metadata.

Artifact	Owns	Must not own
RPLAN	final assignments and plan identity	graph/population context
RCTX	units, graph, population, canonical order	algorithm-specific proof
Certificate	profile checks, verifier results, lineage	legal compliance claim
Manifest	files, hashes, package inventory	substantive validity by itself

Table 1: RPLAN/RCTX fixed-point artifact responsibilities.

The fixed point is a shared final-plan contract, not a new algorithmic claim. Construction, exact optimization, local search, and Pareto comparison may carry different upstream evidence, but their final packages must agree on the same artifact vocabulary.

The atlas framing requires both positive and negative verifier paths to be concrete. A passing package shows the plan, context, certificate, manifest, and hash chain agreeing. A failing package should identify the exact broken invariant: missing context, order mismatch, stale hash, unsupported profile, or broken lineage pointer. That failure reason is part of the audit product rather than an implementation footnote.

Invariant	Required check	Failure mode
Canonical unit order	RPLAN assignments align with RCTX units	order mismatch
Hash integrity	manifest hashes match files	tamper or stale file
Profile scope	certificate names declared checks	unsupported claim
Lineage	algorithm/run ids link to parent artifacts	broken provenance

Table 2: Fixed-point invariants for verifier-facing packages.

3 Audit Semantics

The verifier contract is profile-scoped. It can confirm that declared constraints, lineage, canonical ordering, hashes, and package contents match the profile. It cannot certify fairness, neutrality, or legal sufficiency outside the declared audit profile and available context.

Solver reports, heuristic transcripts, and selected-frontier manifests are upstream evidence. The certificate consumes their final exported artifacts and checks the package under a declared profile. This keeps exact proofs, heuristic-improvement summaries, and Pareto-selection records from being collapsed into one ambiguous claim.

A successful verification should therefore be read as: this plan, with this context, under this profile, with these hashes and lineage records, passed the declared checks. It should not be read as: this plan is fair, legally adequate, or preferred among all available alternatives.

The public verifier workflow is intentionally fixed point shaped: algorithm output becomes a package, the package certificate is checked, and the checked package can then be referenced by reports. The neutral surface is the RPLAN certificate verifier; the BISECT manifest verifier consumes the same package manifest shape as a bridge.

4 Evaluation Plan

Evidence includes RPLAN crate tests, certificate-verifier tests, BISECT bridge tests, final-plan sidecars across construction, exact, local-search, and Pareto families, public golden packages, a U.20 grid10 audit benchmark package, and a tamper/failure-mode catalog.

5 Limitations

Certificates verify a profile, not all law. Some constraints require data that may not be present in every RCTX. External tools must preserve canonical unit ordering, hashes, and

Claim	Current evidence	Publication gap
RPLAN/RCTX separate plan identity from context	rplan-core and rplan-io tests plus public schema examples	larger real-data packages
Certificates verify declared profiles and package integrity	rplan-audit, rplan verify-certificate , public 3x3 package tests, and the grid10 audit benchmark package	Additional manifest/certificate snippets
Algorithm families converge to one final-plan path	Final-plan sidecar tests across T/U papers and public packages by family	larger method-produced packages
Verifier catches tamper and mismatch failures	CLI tests for missing context, plan tamper, context hash changes, unit-order mismatch, profile mismatch, stale context, missing-input results, and lineage-reserved fields	more adversarial fixture variants

Table 3: Claim and evidence ladder for U.20.

package semantics for interoperability to remain valid. The fixed point also depends on honest profile naming: a package can be valid for one profile while lacking the context required by another.

Publication-ready evidence now has an initial public 3x3 package, a grid10 audit benchmark package, and CLI negative tests for missing context, plan tamper, context hash changes, and unit-order mismatch. Additional publication fixtures should still cover stale RCTX context, profile mismatch, unsupported constraint claims, and broken lineage pointers.

6 Conclusion

U.20 is the fixed-point paper for the roadmap. It explains how heterogeneous algorithm families can converge on a common plan/context/audit package while keeping their method-specific evidence and limitations explicit. The reviewed draft now states the invariant directly: certificates are powerful because they are narrow, profile-scoped, and tied to verifiable package semantics rather than broad legal or political conclusions.

References

BISECT project. U.20 plan audit certificates paper plan, 2026. Local planning document and RPLAN schema roadmap.